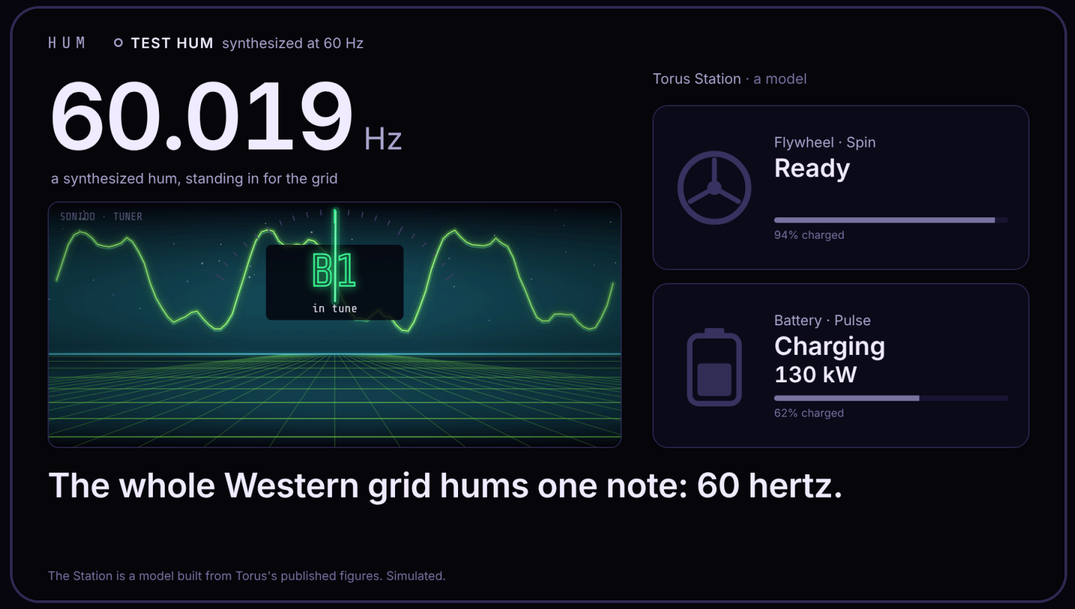
HUM: the power grid, heard through a guitar, and a model of a Torus Station deciding what to do about it.

Every wire in a building hums at the grid's frequency. That one number is shared by the whole Western grid, from Alberta to Baja.

A guitar pickup picks it up. My audio library, sonido, measures it to a thousandth of a hertz and shows it on its tuner.

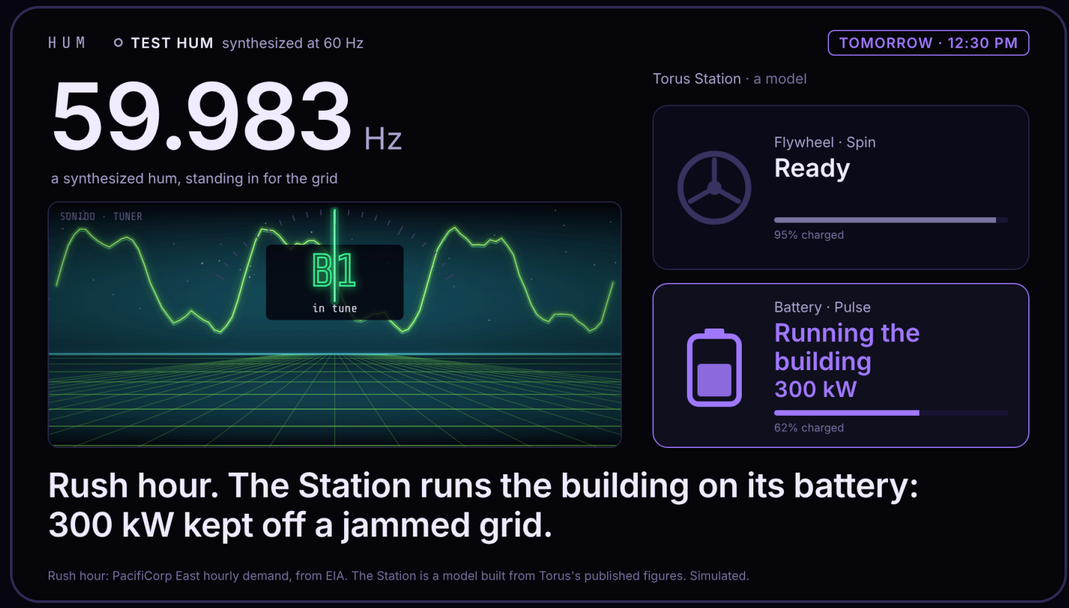


Rush hour you can see coming

The battery runs the building through the jam.

EIA's hourly data for PacifiCorp East shows when the grid gets jammed: this week, from noon to four.

At tomorrow's noon the Station runs the building on its battery, **300 kW kept off the grid**, and recharges when the power is clean.



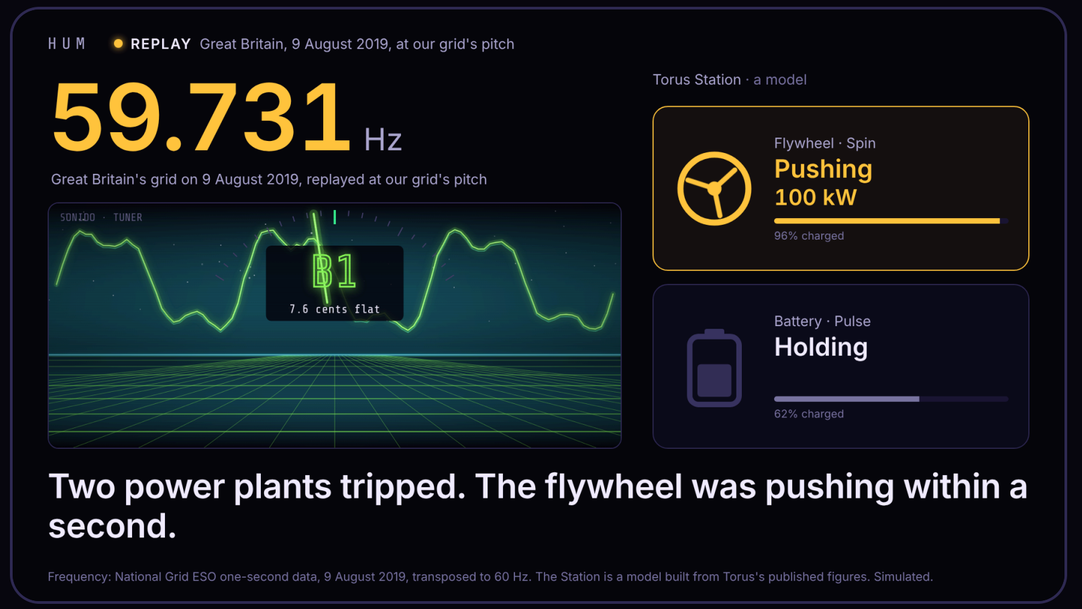
Data: EIA-930, PacifiCorp East. The Station is a model built from Torus's published figures.

A crash you can only hear

The flywheel answers within a second.

On 9 August 2019, two power plants in Britain tripped within a second and a million people lost power. The real frequency record from that day, replayed at our grid's pitch, pulls the number down.

The flywheel pushes at full power before an hourly feed could notice. **When the grid goes dark, the flywheel carries the first seconds and the battery runs the building.**



Data: National Grid ESO one-second frequency, 9 August 2019, transposed to 60 Hz. The Station is a model built from Torus's published figures, simulated.

For your engineers

The software making these decisions can't reach the internet.

It is written in Mentl, a language I built. Its compiler proves this through every function the decision calls. Add one line that reads a price online, and it refuses to build the program.



Replay the demo:
`ampactor.dev/hum`
The code: github.com/ampactor-labs/hum
sonofagoose@pm.me

✕ Refused

One line broke the rule.

```
let price = price_m("https://api.eia.gov/…")
```

This decision is marked as never reaching the internet. Add one line that reads a price online, and the compiler won't build it. Its error names the path the price lookup took.

Thank you, Torus.